

Market Efficiency and Externalities

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Required readings

The core text is N. Gregory Mankiw, *Principles of Microeconomics*.

- **Chapter 7** — Consumers, Producers, and the Efficiency of Markets.
- **Chapter 10** — Externalities.

Two chapters. Read both alongside these slides.

Where we are going

1. **What a buyer will pay** — value becomes a number.
2. **From one buyer to the market** — adding demands sideways.
3. **Consumer surplus** — the buyers' gain, measured.
4. **Producer surplus** — the sellers' gain, measured.
5. **Is the market any good?** — the free market gets its grade.
6. **Costs that spill over** — smoke, bystanders, and too much trade.
7. **Benefits that spill over** — the mirror image: too little.
8. **Fixing it** — taxes, permits, and bargains with no government at all.

PART 1

What a Buyer Will Pay

Grading the market

A market left alone settles where supply meets demand.

Question one: how much good does that do for the people trading in it? Not a feeling — a dollar figure.

Question two: what about people *outside* the trade — the neighbor breathing the smoke, the stranger spared the traffic?

First half: build the measuring instrument. Second half: watch it catch the market failing.

Value becomes a number

Willingness to pay (WTP). The maximum a buyer would hand over for a unit of a good. One cent above it, the buyer walks away.

WTP converts “how much do you want it” into a dollar figure — one per buyer, per unit.

Value is not a mood. It is the price at which you stop.

Four regulars, one cart

A campus coffee cart. Four regulars, one cup each, tops:

Buyer	WTP for a cup
Maya	\$6.00
Omar	\$5.00
Priya	\$3.50
Ben	\$2.00

At $P = \$4.00$: Maya buys ($6.00 \geq 4.00$), Omar buys ($5.00 \geq 4.00$), Priya and Ben pass. $Q^d = 2$.

Price up, count down: that is a demand curve being born.

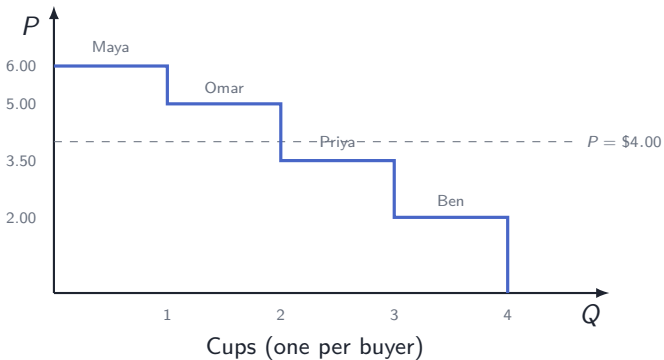
The demand schedule, derived

Buy if $WTP \geq P$. Sweep the price from high to low:

Price range	Who buys	Q^d
above \$6.00	nobody	0
\$5.01 – \$6.00	Maya	1
\$3.51 – \$5.00	Maya, Omar	2
\$2.01 – \$3.50	Maya, Omar, Priya	3
\$2.00 and below	all four	4

Each price cutoff **is** somebody's WTP.

The demand curve is a staircase



The height of the curve at any Q = the WTP of the **marginal buyer** — the one the price is just barely keeping in.

One buyer, several cups

Why does *one person's* demand also slope down? Maya's morning, cup by cup:

Cup	Worth to Maya	Buys at \$4.00?
First	\$6.00	yes
Second	\$4.50	yes
Third	\$3.00	no
Fourth	\$1.50	no

Diminishing marginal utility. Each additional unit adds less satisfaction than the one before, so the dollar value a buyer places on successive units falls.

Two reasons every demand curve falls

As the price drops, quantity demanded rises through two doors:

Across buyers — lower prices recruit people whose WTP was below the old price.

Within each buyer — lower prices justify the next, less-valued unit.

Thousands of buyers $\times$ several units each $\Rightarrow$ the steps shrink until the staircase looks like a smooth line.

A demand curve is every unit anyone wants, ranked best-first.

Active Learning 1

A second cart across campus has five regulars, one cup each:

	Wren	Silas	Nadia	Cole	Tessa
WTP	\$7.00	\$5.50	\$4.50	\$4.00	\$2.50

1. At $P = \$4.50$: who buys, and what is Q^d ?
2. Derive the full demand schedule (price ranges $\rightarrow Q^d$).
3. **The why is the marks:** why does this curve fall in steps, and what, exactly, would smooth it out?

Three minutes. Paper and pencil.

PART 2

From One Buyer to the Market

Write the equations

Two customers, personal weekly demands:

$$q_{\text{Noor}} = 6 - P \qquad q_{\text{Sam}} = 8 - 2P$$

(cups per week; each is zero once P passes the buyer's cutoff: \$6 for Noor, \$4 for Sam).

Read the coefficients:

- Intercepts 6 and 8: cups per week if coffee were free.
- Slopes 1 and 2: cups given up per dollar of price. **Sam is twice as price-sensitive.**

Plug the numbers in

P	Noor: $6 - P$	Sam: $8 - 2P$	Market Q^d
\$6	$6 - 6 = 0$	0 (out)	0
\$5	$6 - 5 = 1$	0 (out)	1
\$4	$6 - 4 = 2$	$8 - 8 = 0$	2
\$3	$6 - 3 = 3$	$8 - 6 = 2$	5
\$2	$6 - 2 = 4$	$8 - 4 = 4$	8
\$1	$6 - 1 = 5$	$8 - 2 = 6$	11

At each price, the market column is the *sum across the row*.

Add quantities, never prices

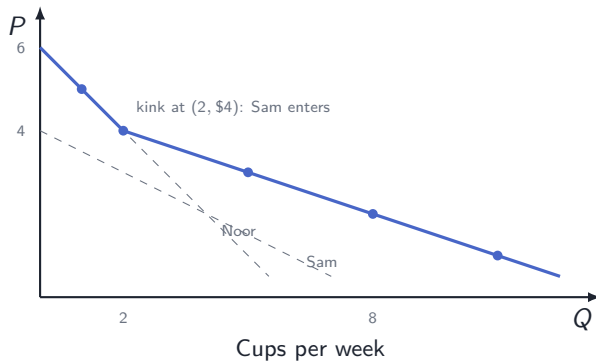
Market demand. At each price, the sum of the quantities all individual buyers demand at that price. Fix P , add the q 's — sideways, never up.

Two price ranges, because Sam quits at \$4:

$$P > 4 : Q^d = 6 - P \qquad P \leq 4 : Q^d = (6 - P) + (8 - 2P) = 14 - 3P$$

Check both ways at $P = 2$: table row $4 + 4 = 8$; equation $14 - 3(2) = 8$. They agree — if they did not, something upstream is wrong.

Draw the graph



Dots from the table; bold line from the equations. They coincide — that is the point.

Scaling up: our market

A campus of thousands of Noors and Sams, summed sideways, gives the market we use all evening:

$$Q^d = 70 - 10P \qquad Q^s = 10P - 10$$

(Q in **thousands** of cups per week.)

Equilibrium:

$$70 - 10P = 10P - 10 \Rightarrow 20P = 80 \Rightarrow P^* = \$4, \quad Q^* = 70 - 10(4) = 30$$

Every number here comes out of these two equations.

Active Learning 2

Two roommates:

$$q_{\text{Ida}} = 5 - P$$

$$q_{\text{Theo}} = 9 - 3P$$

(zero once P passes \$5 and \$3 respectively).

1. Build the market schedule at $P = 4, 3, 2, 1$ — show every substitution.
2. Write the market demand equation for $P > 3$ and for $P \leq 3$.
3. **Check:** get Q at $P = 2$ from the schedule *and* from your equation. If they disagree, find the error.
4. Sketch the market curve and mark the kink.

Five minutes.

PART 3

Consumer Surplus

The buyer's gain has a name

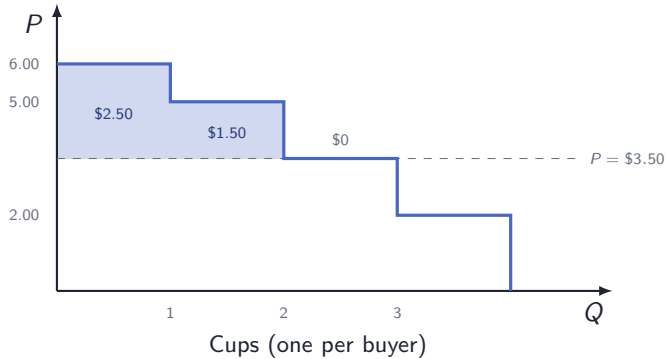
Consumer surplus (CS). What a buyer would have paid minus what the buyer actually pays: $CS = WTP - P$. The buyer's gain from the trade, in dollars.

The four regulars at $P = \$3.50$:

Buyer	WTP	Buys?	$CS = WTP - 3.50$
Maya	\$6.00	yes	$6.00 - 3.50 = \$2.50$
Omar	\$5.00	yes	$5.00 - 3.50 = \$1.50$
Priya	\$3.50	yes	$3.50 - 3.50 = \$0$
Ben	\$2.00	no	—
Total CS			\$4.00

The **marginal buyer** (Priya) gains exactly zero.

Surplus is an area



CS = the area below the demand curve and above the price.

The whole market's CS

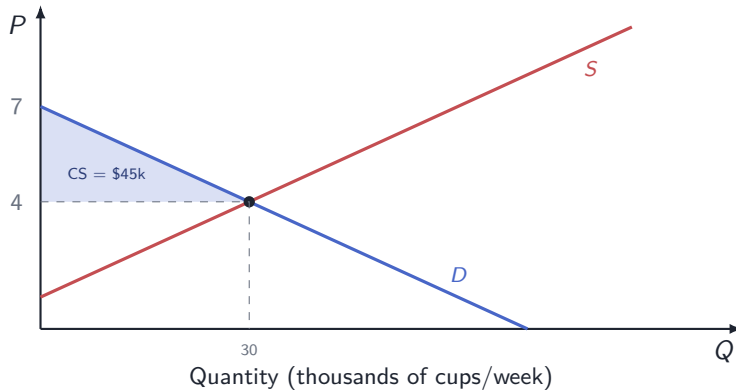
Our market at $P^* = \$4$: height of demand runs from \$7 (the first, most-valued cup) down to \$4 (the marginal cup at $Q = 30$).

Triangle: base = 30, height = $7 - 4 = 3$.

$$CS = \frac{1}{2} \times 30 \times (7 - 4) = \frac{1}{2} \times 30 \times 3 = 45$$

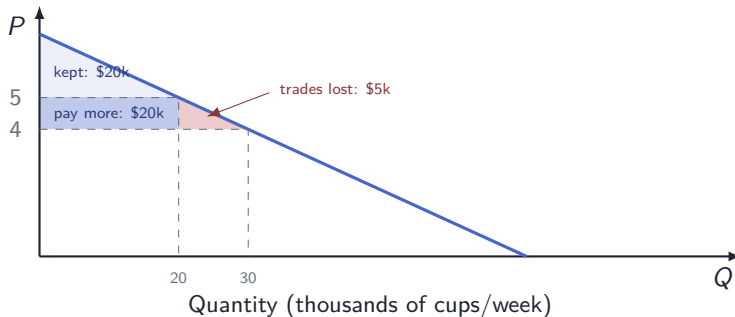
\$45,000 per week of buyer gain — created by the market's mere existence.

The triangle, drawn



A higher price eats CS in two bites

Suppose the price were \$5: $CS = \frac{1}{2} \times 20 \times (7 - 5) = \$20k$ — down \$25k.



The \$25k left in two bites: a rectangle (still buying, paying more) and a triangle (no longer buying).

What CS is actually measuring

If this market vanished tomorrow, its buyers would be exactly **\$45,000 a week poorer** — that is what CS prices.

It is value *as the buyers themselves judge it*: their own WTP, not anyone's opinion of what coffee should be worth.

Next: the sellers' side of the same handshake.

Active Learning 3

Market demand $Q^d = 70 - 10P$. At $P = \$4$, $CS = \$45\text{k}/\text{week}$. Now suppose the price were **\$3**.

1. **Predict before computing:** old buyers save $\$1 \times 30\text{k} = \30k . Will CS rise by *more, less, or exactly* $\$30\text{k}$? Commit, with a reason.
2. Compute CS at $P = \$3$.
3. Split the increase into its two parts.

Four minutes. The prediction counts.

PART 4

Producer Surplus

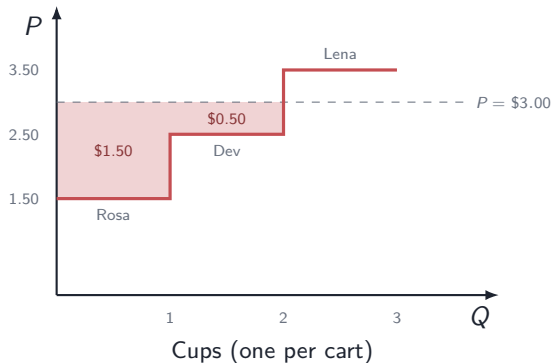
The seller's walk-away price

Cost. Everything a seller gives up to supply one more unit — materials, time, and the next-best use of both (opportunity cost). Cost is the seller's walk-away price: the lowest price at which the unit is worth supplying.

Three carts, cost per cup:

Cart	Cost per cup	Sells at $P = \$3.00$?
Rosa	\$1.50	yes
Dev	\$2.50	yes
Lena	\$3.50	no

Supply is a staircase too



Height of the supply curve at any Q = the **cost of the marginal seller**.

Producer surplus, defined and scaled

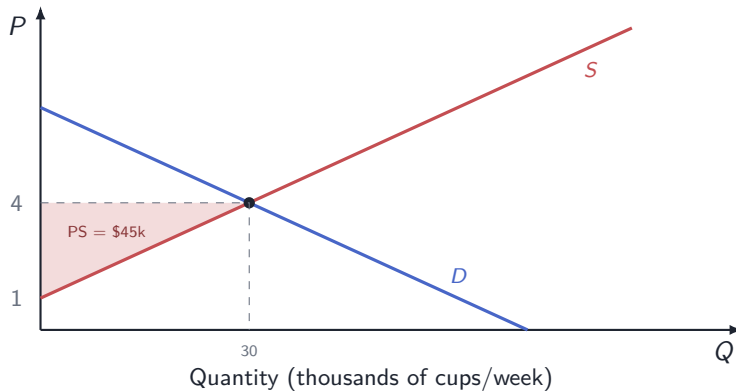
Producer surplus (PS). What a seller is paid minus the seller's cost: $PS = P - \text{cost}$. The area below the price and above the supply curve.

Our market at $P^* = \$4$: supply height runs \$1 (cheapest cup) up to \$4 (marginal cup at $Q = 30$).

$$PS = \frac{1}{2} \times 30 \times (4 - 1) = \frac{1}{2} \times 30 \times 3 = 45$$

\$45,000 per week of seller gain. The marginal seller — like the marginal buyer — gains zero.

The sellers' triangle



Two triangles, one symmetry

	Consumer surplus	Producer surplus
Private number	WTP (max they'd pay)	cost (min they'd take)
Gain per unit	$WTP - P$	$P - \text{cost}$
On the diagram	below D , above price	above S , below price
At the margin	zero (marginal buyer)	zero (marginal seller)
Our market	\$45k/week	\$45k/week

Equal here by construction of the example — not a law. The symmetry of *logic* is the law.

Active Learning 4

Supply $Q^s = 10P - 10$. At $P = \$4$, $PS = \$45\text{k}/\text{week}$.

1. In one sentence: *why* is the marginal seller's surplus exactly zero?
2. Suppose the price were **\$5**. Compute PS.
3. Split the increase into its two parts (old units, new units).

Three minutes. Mirror everything you did for buyers.

PART 5

Is the Market Any Good?

Add the gains: total surplus

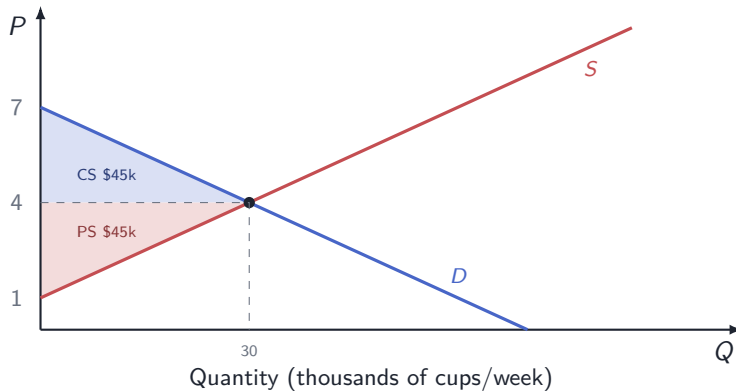
Total surplus (TS). Consumer surplus plus producer surplus: the entire gain the market creates for its participants.

$$TS = \underbrace{(\text{value to buyers} - \text{paid})}_{\text{CS}} + \underbrace{(\text{received} - \text{cost})}_{\text{PS}} = \text{value to buyers} - \text{cost to sellers}$$

The money paid **cancels** — it changed hands, it didn't vanish.

Our market: $TS = 45 + 45 = \$90,000$ per week.

Ninety thousand dollars a week, drawn



The whole wedge between D and S , from 0 to Q^* .

The right buyers, the right sellers

At $P^* = \$4$, with no one directing anything:

- Every buyer with $WTP \geq \$4$ buys — the 30,000 **highest-value** cups are the ones consumed.
- Every seller with $\text{cost} \leq \$4$ sells — the 30,000 **cheapest** cups are the ones produced.

Reassign any cup from a buyer to a non-buyer, or a seller to a non-seller, and total surplus **falls**.

The price is a sorting machine. Nobody built it; nobody runs it.

The right amount

Value of the marginal cup: $(70 - Q)/10$. Its cost: $(Q + 10)/10$.

Q (000s)	Marginal value	Marginal cost	Gap
20	$\frac{70-20}{10} = 5.00$	$\frac{20+10}{10} = 3.00$	+2.00
25	$\frac{70-25}{10} = 4.50$	$\frac{25+10}{10} = 3.50$	+1.00
30	$\frac{70-30}{10} = 4.00$	$\frac{30+10}{10} = 4.00$	0.00
35	$\frac{70-35}{10} = 3.50$	$\frac{35+10}{10} = 4.50$	-1.00
40	$\frac{70-40}{10} = 3.00$	$\frac{40+10}{10} = 5.00$	-2.00

Below 30: stopping early **forgoes** gains. Past 30: each extra cup **destroys** value.

TS peaks at $Q = 30$ — the market's own unsupervised choice.

Efficiency. An allocation is efficient when it maximizes total surplus: no change in who buys, who sells, or how much is traded can create more value.

What efficiency does **not** say: nothing about *who gets* the surplus.

Equality — how the gains are shared — is a separate question, and a *normative* one. Our instrument certifies the size of the pie, not the fairness of the slices.

Adam Smith saw it in 1776

“It is not from the benevolence of the butcher, the brewer, or the baker that we expect our dinner, but from their regard to their own interest. ... [Each] is led by an invisible hand to promote an end which was no part of his intention.”

— Adam Smith, *The Wealth of Nations* (1776)

In our language: **self-interest, filtered through a market price, maximizes total surplus.**

We just proved with a table what Smith could only name.

The fine print

That verdict leaned on two quiet assumptions:

1. **Competition.** No single seller or buyer sets the price. (When this fails — market power — is a story for later in the material.)
2. **Everyone affected is at the table.** The trade touches nobody but its own buyer and seller.

Market failure. A situation in which an unregulated market's equilibrium fails to maximize total surplus.

From here on: assumption 2 breaks.

Active Learning 5

The mayor, “to keep coffee special,” will license only **20 thousand** cups per week (price free to adjust).

1. **Predict:** does total surplus rise, fall, or hold at \$90k?
2. Compute TS at $Q = 20$ **two independent ways**:
 - (a) \$90k minus the lost triangle;
 - (b) directly: gain per cup is $6 - 0.2Q$ dollars (it is \$6.00 on the first cup, \$2.00 on the 20,000th) — average it over 20k cups.
3. Why *must* (a) and (b) agree — and what would you conclude if they did not?

Five minutes.

PART 6

Costs That Spill Over

The bystander

Externality. A cost or benefit from a trade that lands on someone who is not a party to it — a bystander, neither paying nor paid.

Negative — harm spills out:

- wood smoke drifting downhill
- a gas leaf blower at 7 a.m.
- one more SUV at rush hour

Positive — benefit spills out:

- a rain garden soaking up street runoff
- a neighbor's well-kept facade
- every commuter who bikes instead

The market's books only ever record the two people shaking hands.

A mountain town that burns wood

The market for firewood bundles in a valley town:

$$Q^d = 100 - 10P \qquad Q^s = 10P - 20$$

(Q in **hundreds** of bundles per week.)

Equilibrium:

$$100 - 10P = 10P - 20 \Rightarrow 20P = 120 \Rightarrow P^* = \$6, \quad Q^* = 100 - 10(6) = 40$$

By everything so far: efficient. **For buyers and sellers.**

But every bundle burned sends smoke down the valley.

What a bundle truly costs

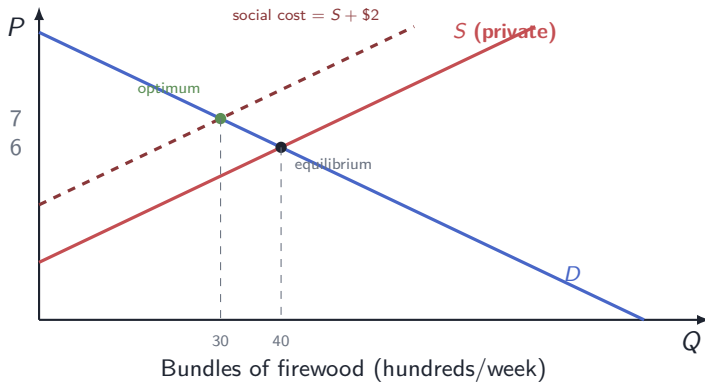
Suppose the smoke from each bundle does **\$2.00** of harm to bystanders (health, laundry, cleaning).

Social cost. Private cost plus external cost: what one more unit costs *society*, counting the bystanders.

Q	WTP	Private cost	Social cost	Society's gain
20	$\frac{100-20}{10} = 8.00$	$\frac{20+20}{10} = 4.00$	$4.00 + 2 = 6.00$	+2.00
30	$\frac{100-30}{10} = 7.00$	$\frac{30+20}{10} = 5.00$	$5.00 + 2 = 7.00$	0.00
40	$\frac{100-40}{10} = 6.00$	$\frac{40+20}{10} = 6.00$	$6.00 + 2 = 8.00$	-2.00

The market stops where *private* books balance ($Q = 40$). Society's books balance at $Q = 30$.

Equilibrium vs. optimum, drawn



With a negative externality, the market **overshoots**: $Q^* = 40 > 30$.

The thousand bundles too many

Bundles between the optimum and the equilibrium, one at a time:

Bundle (00s)	Worth to buyer	Social cost	Society loses
32	$\frac{100-32}{10} = 6.80$	$\frac{32+40}{10} = 7.20$	0.40
36	$\frac{100-36}{10} = 6.40$	$\frac{36+40}{10} = 7.60$	1.20
40	$\frac{100-40}{10} = 6.00$	$\frac{40+40}{10} = 8.00$	2.00

Buyer and seller both said *yes* to every one of these — their private gain is real. The \$2 landing on the bystanders outweighs it.

Each of these trades *destroys* value — it just destroys it off the books.

Name the damage: deadweight loss

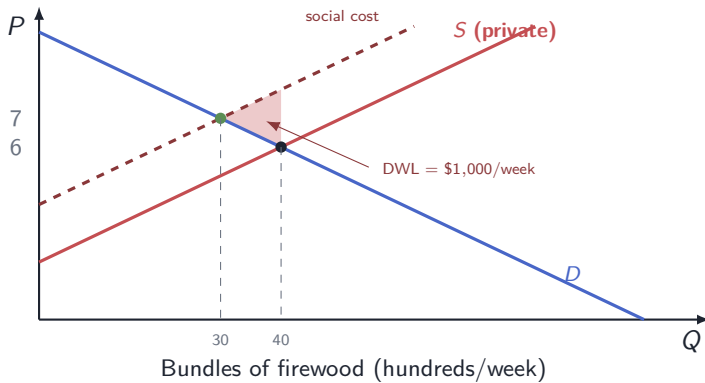
Deadweight loss (DWL). Total surplus lost because the quantity traded is not the efficient quantity — value destroyed by trading too much or too little, collected by no one.

Here: losses on the extra bundles grow from \$0 (at 3,000) to \$2.00 (at 4,000) — a triangle:

$$DWL = \frac{1}{2} \times 10 \times \$2.00 = 10 \Rightarrow \$1,000 \text{ per week}$$

Too much trade destroys value as surely as too little. The triangle measures either.

The triangle, drawn



Make the market see the smoke

Internalizing an externality. Restructuring incentives so the people in the trade face the full social cost (or value) of it.

A **corrective tax** (a *Pigovian* tax) of \$2.00 per bundle — set equal to the external cost — charged to sellers:

$$Q^s : 10(P - 2) - 20 = 10P - 40 \quad 100 - 10P = 10P - 40 \Rightarrow P = \$7, \quad Q = 30$$

Buyers pay \$7; sellers keep \$5; the town collects $\$2 \times 3,000 = \$6,000/\text{week}$.

$Q = 30$: the taxed market lands *exactly* on the social optimum. This tax removes a deadweight loss.

Active Learning 6

New measurements: the smoke harm is **\$4.00** per bundle, not \$2.00. (Market: $Q^d = 100 - 10P$, $Q^s = 10P - 20$.)

1. **Predict before computing:** which way does the social optimum move? Does the *unregulated equilibrium* move at all? Why?
2. Compute the new optimum, and the corrective tax that lands the market on it. Give both prices (buyers pay, sellers keep) and **check the quantity two ways** (demand side, supply side).
3. Compute the DWL of leaving the market alone.

Five minutes. The why in Q1 is the marks.

PART 7

Benefits That Spill Over

Every ride is a car that isn't there

E-bike share in a gridlocked city:

$$Q^d = 80 - 10P$$

$$Q^s = 10P - 20$$

(Q in **hundreds** of rides per day.)

Equilibrium:

$$80 - 10P = 10P - 20 \Rightarrow 20P = 100 \Rightarrow P^* = \$5, \quad Q^* = 80 - 10(5) = 30$$

Every rush-hour ride is one car *not* in everyone else's way.

What a ride is truly worth

The bystanders' gain — less congestion, less exhaust, for drivers who never touch a bike: **\$2.00 per ride.**

Social value. Private value (WTP) plus external benefit: what one more unit is truly worth to society.

The rider weighs the fare against *their own* benefit only — the \$2 enters **nobody's** decision.

Mirror image of the smoke: a benefit the market never sees.

The rides that don't happen

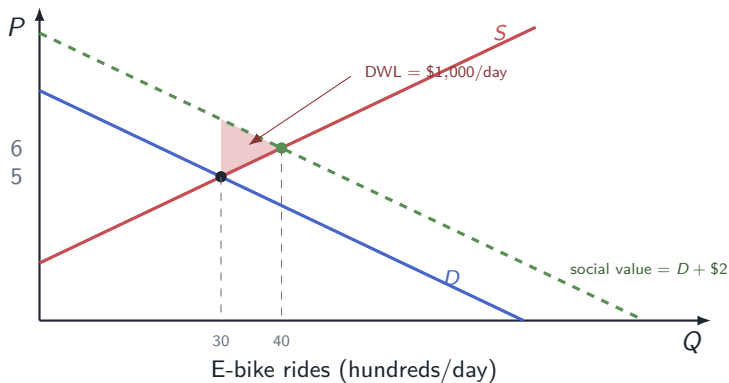
$$\text{Social value} = \frac{100-Q}{10}; \quad \text{cost} = \frac{Q+20}{10}:$$

Rides (00s)	Social value	Cost	Society's gain
30	$\frac{100-30}{10} = 7.00$	$\frac{30+20}{10} = 5.00$	+2.00
35	$\frac{100-35}{10} = 6.50$	$\frac{35+20}{10} = 5.50$	+1.00
40	$\frac{100-40}{10} = 6.00$	$\frac{40+20}{10} = 6.00$	0.00
45	$\frac{100-45}{10} = 5.50$	$\frac{45+20}{10} = 6.50$	-1.00

Optimum: 4,000 rides. Market: 3,000. The missing thousand? The rider pays the whole fare; the bystanders' \$2 gain is priced at zero.

With a positive externality, the market *undershoots*.

The triangle returns



Missing trades, 30–40: value grows $\$2.00 \rightarrow \0 . $DWL = \frac{1}{2} \times 10 \times \$2 = \$1,000/\text{day}$.

Pay the spillover: a subsidy

Give riders **\$2.00 off** per ride — equal to the external benefit. Demand as sellers see it:

$$Q^d = 80 - 10(P - 2) = 100 - 10P \quad 100 - 10P = 10P - 20 \Rightarrow P = \$6, \quad Q = 40$$

Operators receive \$6; riders pay $6 - 2 = \$4$; city pays $\$2 \times 4,000 = \$8,000/\text{day}$.

The outlay is a **transfer** (taxpayers $\rightarrow$ riders/operators), not a loss. What society *gains*: the \$1,000/day triangle, back.

Tax what harms bystanders. Subsidize what helps them. Same logic, opposite signs.

The spillover scorecard

	Market vs. optimum	The triangle sits	Remedy
Negative ext.	trades too much	between D and social cost	tax = external cost
Positive ext.	trades too little	between social value and S	subsidy = external benefit

The market fails in the direction of whatever it cannot see.

Active Learning 7

For each: **positive**, **negative**, or **no** externality? Name the bystander. One reason each.

1. A gas leaf blower at 7 a.m.
2. A homeowner's rain garden that soaks up street runoff.
3. Repainting your own bedroom.
4. A speakerphone call in the quiet car.
5. Rooftop solar that trims the grid's strained peak load.
6. One more heavy SUV joining rush hour.

Then, for #1 and #5: over- or under-provided, and — tax or subsidy?

Three minutes.

PART 8

Fixing It, With and Without Government

Two toolkits, and a third way

Command-and-control: forbid or require behavior directly — emission caps per firm, mandated equipment, permitted hours.

Market-based: put a price on the spillover and let people choose — corrective taxes and subsidies; *tradable permits*.

No government at all: sometimes the neighbors fix it themselves. (Last stop.)

The test for any tool: does it get the quantity right — at the lowest cost of getting it right?

Two factories, one river

Northside Mill and Harbor Chemical each discharge **80 tons/month** into the same river. The town wants the total down to **100**.

Cost of cutting one ton: **Mill \$50 — Harbor \$150**.

Rule: “each firm cuts 30.”

	Tons cut	Cost
Northside Mill	30	$30 \times \$50 = \$1,500$
Harbor Chemical	30	$30 \times \$150 = \$4,500$
Total	60	\$6,000/month

The rule hits the quantity — but it is **blind to who can cut cheaply**.

Sell the right to pollute

Tradable pollution permits. Each permit licenses one ton of discharge. Total permits issued = the cap. Firms may buy and sell them.

Issue 100 permits, 50 to each firm. They trade at \$100:

	Emits	Cuts	Net cost
Mill (sells 30)	20	60	$60 \times 50 - 30 \times 100 = \0
Harbor (buys 30)	80	0	$30 \times 100 = \$3,000$
Total	100	60	\$3,000/month — half the rule.

Same river — but the **cutting migrated to the cheap cutter**. Society's real bill: the Mill's \$3,000 of abatement; the permit payments are transfers.

Tax or permits: price vs. quantity

Both put a **price on a ton of harm**:

- **Corrective tax** — fixes the *price* of harming; quantity finds itself.
- **Permits** — fix the *quantity* of harm; the price finds itself.

Either beats a uniform rule when firms differ — and firms always differ.

Price the harm; let the market find who abates.

No government: strike a deal

Café Solana's Friday live-music night. The noise reaches Harriman Books next door.

worth to café: \$400/wk

costs bookshop: \$600/wk

The café has the legal right to play.

The bookshop offers \$500 for quiet Fridays. Café accepts ($500 > 400$): music stops. Bookshop keeps $600 - 500 = \$100$; café gains $500 - 400 = \$100$.

Efficient outcome ($600 > 400$: quiet is worth more) — reached by a private check, no regulator in sight.

(If music were worth \$800 to the café: no offer the bookshop can afford tempts it. Music plays — *also* efficient.)

The Coase theorem — and its limits

Coase theorem. If the affected parties can bargain at no cost, they reach the efficient outcome on their own — *whichever side the law initially favors*.

Flip the right (a quiet ordinance, music worth \$800): the *café* now pays (\$600–\$800 works) and plays on. **The right decides who pays whom, not what happens.**

Why bargaining fails at scale:

- transaction costs — lawyers cost more than the stakes;
- breakdowns — each side holds out for the larger slice;
- numbers — 5,000 households cannot bargain with every stove.

Two neighbors can Coase. A town cannot. That is when policy earns its keep.

Active Learning 8

Two plants, cutting costs **A: \$80/ton**, **B: \$30/ton**. Each emits 50 tons; the state wants the total down to **60** (cut 40).

1. Cost of the rule “each plant cuts 20.”
2. 60 permits issued, 30 each, trading at **\$55**: who sells, who buys, each plant’s net cost, and **society’s real abatement bill**.
3. **The reasoning is the marks**: for what *range* of permit prices are both plants better off than under the rule? Why are the endpoints exactly \$30 and \$80?

Five minutes.

The market maximizes the gains of the people it can see.

Buyers and sellers: every last dollar, no planner required.

Bystanders: invisible — so it trades too much smoke
and too few good deeds, until a price makes it look.